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Chinese Journal of Mechanical Engineering (2025) 38:78
https://doi.org/10.1186/s10033-025-01235-5

Chinese Journal of Mechanical Engineering

ORIGINAL ARTICLE

Open Access

Performance Analysis and Multi-Objective Optimization of Functional Gradient Honeycomb Non-pneumatic Tires

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Abstract

The spoke as a key component has a significant impact on the performance of the non-pneumatic tire (NPT). The current research has focused on adjusting spoke structures to improve the single performance of NPT. Few studies have been conducted to synergistically improve multi-performance by optimizing the spoke structure. Inspired by the concept of functionally gradient structures, this paper introduces a functionally gradient honeycomb NPT and its optimization method. Firstly, this paper completes the parameterization of the honeycomb spoke structure and establishes the numerical models of honeycomb NPTs with seven different gradients. Subsequently, the accuracy of the numerical models is verified using experimental methods. Then, the static and dynamic characteristics of these gradient honeycomb NPTs are thoroughly examined by using the finite element method. The findings highlight that the gradient structure of NPT-3 has superior performance. Building upon this, the study investigates the effects of key parameters, such as honeycomb spoke thickness and length, on load-carrying capacity, honeycomb spoke stress and mass. Finally, a multi-objective optimization method is proposed that uses a response surface model (RSM) and the Non-dominated Sorting Genetic Algorithm - II (NSGA-II) to further optimize the functional gradient honeycomb NPTs. The optimized NPT-OP shows a 23.48% reduction in radial stiffness, 8.95% reduction in maximum spoke stress and 16.86% reduction in spoke mass compared to the initial NPT-1. The damping characteristics of the NPT-OP have also been improved. The results offer a theoretical foundation and technical methodology for the structural design and optimization of gradient honeycomb NPTs.

Keywords Non-pneumatic tires, Honeycomb structure, Gradient structure, Multi-objective optimization

1 Introduction

Automobile tires, integral components of vehicles, are sophisticated composite material structures [1]. Serving as a crucial element in the vehicle's propulsion system, the tire is the only component that interacts with the road. Its primary functions include bearing loads, generating ground forces, and mitigating ground disturbances [2]. Tires play a pivotal role in influencing vehicle handling and stability, ride comfort, and overall performance. For decades, traditional pneumatic tires have dominated the global tire market due to their advantages of low contact pressure, low energy loss, low mass, and low vertical stiffness [3]. However, traditional pneumatic tires pose potential risks, including those related to air leakage, unstable pressure, blowouts, and wear. According to statistics, 46% of highway traffic accidents were caused by tire hazards, with flat tires accounting for a staggering 70% of accidents [4]. Therefore, to ensure vehicle safety, tire manufacturers and relevant research institutes have been seeking practical and viable tire safety technologies. Tire safety technology is a comprehensive technology

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that not only maintains the basic functions of a tire but also enhances safety or eliminates tire hazards altogether. Current technological approaches for improving tire safety involve either enhancing the safety of pneumatic tires or adopting nonpneumatic tires (NPTs). Pneumatic safety tires are currently used as emergency safety measures and are not suitable for prolonged driving. Therefore, research on NPT technology has received much attention in recent years.

The primary distinction between NPTs and traditional pneumatic tires is that NPTs use a support structure instead of inflation pressure. A typical NPT comprises three main components: the tire body, rim, and support structure. The support structure, a crucial element, absorbs impact and provides flexibility to the wheels under static and dynamic conditions [5]. Over several decades of research, various types of NPTs have been developed, notable examples being Tweel NPTs, mechanical elastic NPTs (MENPTs), and honeycomb NPTs. Tweel tires exhibit performance characteristics similar to those of pneumatic tires, making them suitable replacements in specific operating conditions [6–8]. The support structure of MENPTs, crafted from highly elastic metal, has an excellent ability to absorb impacts and vibrations [9]. Umesh et al. highlighted that honeycomb spoke tires deliver uniform traction and have low wear performance, making them a viable alternative to traditional pneumatic tires [10]. The support structure of NPTs has a significant impact on the mechanical properties of the wheels. NPTs typically feature multiple specially designed spokes that must meet specific stiffness and elasticity requirements under alternating tensile and compressive loads, which presents a considerable challenge in the selection of spoke materials and structural design.

Thus far, researchers and wheel manufacturers have proposed diverse design solutions for the spoke structure of NPTs and have conducted comprehensive research on the effects of various structural parameters on mechanical performance. Honeycomb structures exhibit high stiffness and strength in the out-of-plane direction, coupled with relatively low mechanical resistance and high elasticity in the in-plane direction. Therefore, honeycomb structures are considered to have considerable application potential in NPTs [11]. Vinay et al. applied the honeycomb structure to the design of NPTs and revealed that slight changes in honeycomb shape could result in notable changes in tire radial stiffness. Moreover, increasing the thickness of honeycomb spokes was found to diminish the deformation of the spokes [12]. Aboul-Yazid et al. compared the effect of different spoke structures on tire contact pressure, finding that the honeycomb spoke structure can reduce contact pressure and stress on the NPT spokes [13]. The honeycomb spoke structure is composed of polygonal cells, and changing the arrangement, wall thickness, or length of the cells can affect the overall performance characteristics of the NPTs. Therefore, it is necessary to conduct in-depth research on the relationship between the geometric structure of honeycomb spokes and their mechanical properties. Ganniari-Papageorgiou et al. [14] conducted a comprehensive investigation of the effects of honeycomb structure design parameters on the performance of NPTs. Their findings revealed that the density, thickness, and internal angle of honeycomb cells have a significant impact on the vertical stiffness. Moreover, a reduction in the length of honeycomb cells or an increase in cell density can significantly reduce internal stress. In conventional honeycomb structures, where hexagonal cells are uniform in size and arranged in rows and columns, energy absorption is confined to a limited range. Conversely, nonuniform structures can absorb energy over a wider range [15, 16]. Liang et al. demonstrated that honeycomb structures offer substantial advantages in load-bearing and energy absorption [17, 18]. Tao et al. introduced a grading design for honeycomb structures, which increased the specific energy (SEA) by 70% compared to conventional honeycomb structures [19]. Sahu et al. used three-dimensional (3D) printing technology to manufacture graded honeycomb structure samples and conducted cyclic compression and free vibration experiments. Their experimental results indicated that the hybrid gradient honeycomb structure exhibits high damping capacity and serves as an effective energy absorption element [20]. Thus, the gradient honeycomb structure demonstrates superior mechanical properties, making it a promising support structure for NPTs.

There is a wide variety of support structures for NPTs. To seek the best structural parameters, the optimization research of NPTs has been progressing in recent years. Veeramurthy et al. found that shear modulus and shear band thickness of polyurethane are the most important design parameters affecting rolling resistance, vertical stiffness, and contact pressure of NPTs. They used RSM to find the optimum design values to minimize the rolling resistance [21]. Xiao et al. proposed a virtual test field modeling method based on finite element analysis. Accurate life prediction of a MENPT was performed using this method. An approximate RSM applicable to MENPTs was established. On this basis, the optimization of tire life was carried out for MENPTs [22]. Wang et al. investigated the effects of key spoke structural parameters, such as curvature, array period and thickness, on the fatigue durability of NPTs by numerical simulation methods. An optimization method based on the Kriging model and NSGA-II was proposed to complete the structural optimization of the BP-NPTs. The better fatigue life of the

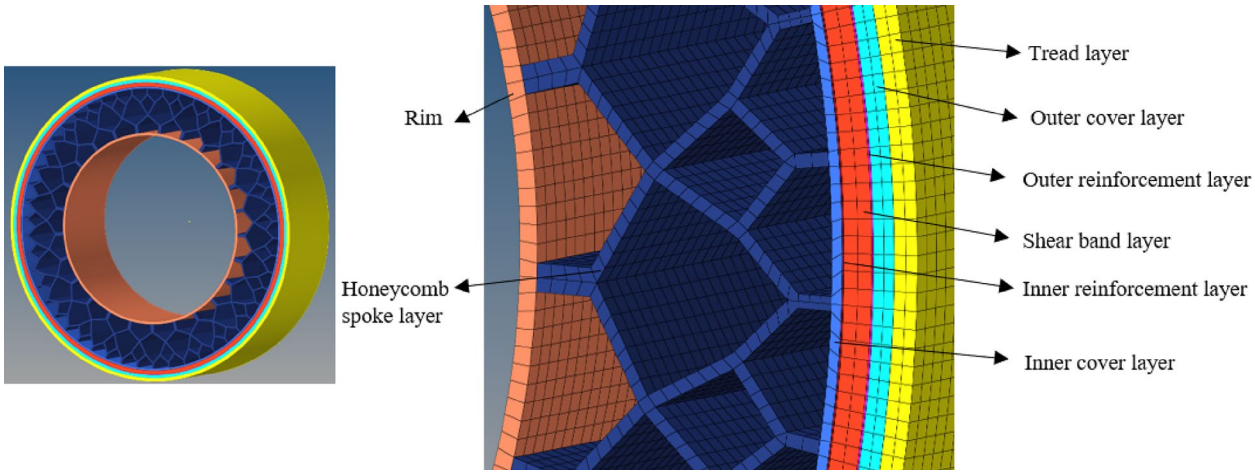


Figure 1 Structural diagram of the NPT with honeycomb spokes

BP-NPTs was finally obtained [23]. Current research on the NPT structure optimization focuses on optimizing resistance and fatigue life. Parametric modeling of spoke structures is also rarely implemented. This increases the difficulty of the research. There are also fewer studies related to the structural optimization of honeycomb NPTs. This paper is dedicated to parameterizing the honeycomb spoke structure and accomplishing multi-objective optimization of honeycomb NPTs. However, the design of honeycomb structures is inherently complex, especially when introducing grading and gradient concepts and when attempting to meet the mechanical requirements of NPTs. This complexity considerably increases the difficulty of designing NPTs with a honeycomb structure.

Existing designs of NPTs with a honeycomb structure struggle to achieve parameterization, and multi-objective optimization methods are rarely used [24]. This paper addressed these challenges by introducing gradient concepts in the design of honeycomb-structured NPTs, along with achieving parameterization. Through the design of the experiment (DOE) and structural parameters research, the effects of various parameters, such as honeycomb spoke length and thickness, on radial stiffness, maximum stress of the spokes, and other static and dynamic performance aspects of NPTs were thoroughly examined. Furthermore, multi-objective optimization design for honeycomb NPTs was achieved through RSM and the NSGA-II. This article is organized as follows: Section 2 introduces the simulation models, encompassing geometric parameters, material properties, and finite element analysis methods. Section 3 presents an investigation of the static and dynamic performance of gradient honeycomb NPTs. Section 4 focuses on multi-objective optimization based on the findings from research on the gradient honeycomb NPTs. Finally, Section 5 presents a summary of the research results.

2 Simulation Model

2.1 Geometrical Parameters

The 3D models of the NPTs were initially established using the commercial NX UG software, and their corresponding finite element (FE) models were developed using the HyperMesh software. A honeycomb NPT comprises a tread layer, an outer cover layer, a shear band layer, two reinforcement layers, an inner cover layer, a honeycomb spoke layer, and a rim. The basic composition of a honeycomb NPT is presented in Figure 1. The specifications of the NPTs were designed with reference to 215/60 R17 pneumatic tire. The specific structural parameters are presented in Table 1.

Honeycomb spokes play a crucial role as supporting and energy-absorbing structures in the NPTs. The structural design of these honeycomb spokes has a considerable effect on the various properties of the NPTs. The honeycomb structure of the NPTs is depicted in Figure 2. The in-plane effective moduli of the honeycomb structures were determined according to the honeycomb material theory [25–30], which are given as follows:

$$E_r^* = E_s\left(\frac{a}{l}\right)^3 \frac{\cos\theta}{(h/l+\sin\theta)\sin^2\theta}, \tag{1}$$

$$E_r^* = E_s\left(\frac{a}{l}\right)^3 \frac{\cos\theta}{(h/l+\sin\theta)\sin^2\theta}, \tag{2}$$

$$G_{r\Omega}^* = E_s\left(\frac{a}{l}\right)^3 \frac{(h/l+\sin\theta)}{(h/l)^2(1+2h/l)\cos\theta}, \tag{3}$$

Table 1 Dimensions of the NPT components, excluding the honeycomb spoke layer

	Tread	Outer cover	Outer reinforcement	Shear band	Inner reinforcement	Inner cover	Rim
Radius (mm)	337	330	329.36	320	319.36	315.90	210
Thickness (mm)	7.90	7	0.64	9.36	0.64	3.46	5.90

where E_r^*, E_Ω^* and $G_{r\Omega}^*$ are the effective moduli of the 2D spoke in the radial, circumferential and shear directions, respectively. Moreover, h equals $l_3\cos\angle 2$, where l denotes the length of spoke l_2 and l_4, and θ represents angles taken as $\angle 3$ and $\angle 4$. E_s denotes the Young's modulus of the base material of the spokes, and a is considered the corresponding spoke thickness. It needs to be noted that the thickness dimensions of the walls of each spoke may be different. The effective moduli corresponding to spoke l_2 and l_4 are then averaged to obtain the in-plane effective modulus of the honeycomb structure.

Based on Eqs. (1)–(3), it can be deduced that the mechanical properties of honeycomb NPTs are related to the length, angle, and thickness of the honeycomb spokes.

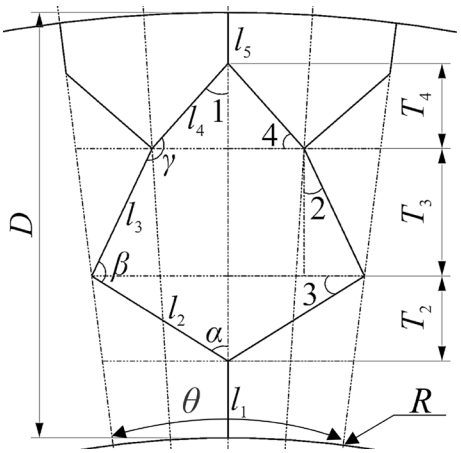


Figure 2 Diagram of honeycomb spoke structure

2.2 Parameterization of the Honeycomb Spoke Structure

The structure of an individual honeycomb cell is illustrated in Figure 2. The complete honeycomb spoke support structure of the NPTs was derived from a circular array of 25 individual honeycomb cells around the center of the wheel. Based on the geometrical structural relationships in Figure 2, the lengths of l_2, l_3, and l_4 in the honeycomb structure and their angles with the vertical diameter can be determined as follows:

$$\alpha = \arctan \frac{(R + l_1 + T_2)\tan\frac{\theta}{2}}{T_2}, \tag{4}$$

$$l_2 = \left[(R + l_1 + T_2)\tan\frac{\theta}{2}\right] / \sin\alpha, \tag{5}$$

$$\angle 2 = \arctan \frac{(R + l_1 + T_2)\tan\frac{\theta}{2} - (R + l_1 + T_2 + T_3)\tan\frac{\theta}{4}}{T_3}, \tag{6}$$

$$l_3 = T_3 / \cos\angle 2, \tag{7}$$

$$\angle 1 = \arctan \frac{(R + l_1 + T_2 + T_3)\tan\frac{\theta}{4}}{T_4}, \tag{8}$$

$$l_4 = \left[(R + l_1 + T_2 + T_3)\tan\frac{\theta}{4}\right] / \sin\angle 1, \tag{9}$$

where R is the outer radius of the rim; θ is the angle of the center of the circle corresponding to an individual honeycomb cell; and T_2, T_3, and T_4 are the projected lengths of spokes l_2, l_3, and l_4, respectively, in the vertical diameter direction.

The length of each edge of the honeycomb spoke is a key parameter that alters the honeycomb structure and influences the performance of the NPTs. According to Eqs. (4)–(9), the length of each edge can be determined by projecting it in the radial direction, thus quantifying the distinct structure of the honeycomb spokes. By incorporating the specific thickness parameters of each spoke, the entire honeycomb support structure can be determined. The initial parameters of the honeycomb structure are detailed in Table 2.

To investigate the effect of the honeycomb hexagonal structure and the gradient in honeycomb thickness on NPT performance, while considering computational cost, the length of l_5 and l_1 were kept constant at 12 and 18 mm, respectively. Similarly, the thickness of l_5 and l_1 were kept constant at 4 and 8 mm, respectively. The focus was on studying the effects of variations in the shape and thickness gradient of the honeycomb central hexagonal structure on NPT performance. To construct the

Table 2 Specific parameters of the honeycomb structure

	D	l_5	T_4	T_3	T_2	l_1
Length (mm)	100	12	20	30	20	18

Table 3 Spoke thickness parameters of hybrid gradient honeycomb NPTs

Plans	NPT-1	NPT-2	NPT-3	NPT-4	NPT-5	NPT-6	NPT-7
l_4 thickness (mm)	3	3	4	4	4	5	5
l_3 thickness (mm)	4	5	3	4	5	3	4
l_2 thickness (mm)	5	4	5	4	3	4	3
Mass (kg)	29.57	29.54	29.79	29.74	29.70	29.97	29.90

Table 4 Material properties of the NPTs [31]

Materials	Density (kg/m^3)	Young's modulus (GPa)	Shear modulus (GPa)	Poisson's ratio	Yield strength (MPa)
Aluminum alloy (Al 7075-T6)	2800	72.0	27.0	0.33	503
High-strength steel (ANSI 4340)	7800	210.0	76.0	0.29	1240
Polyurethane	1100	32.0	10.8	0.49	140
Synthetic rubber	1043	11.9	4.0	0.49	16

Table 5 Parameters of polyurethane and synthetic rubber material [31]

i	Polyurethane				Synthetic rubber			
	μ_i (MPa)	a_i	g_i	τ_i	μ_i (MPa)	a_i	g_i	τ_i
1	13.546	1.513	0.125	0.002	13.356	1.633	0.200	0.002
2	− 2.338	2.212	0.125	0.020	− 6.631	1.900	0.200	0.020
3	0.093	− 2.471	0.125	0.200	0.058	− 2.456	0.200	0.200

gradient honeycomb structure and explore the effect of spoke thickness on NPT performance, this study selected thicknesses of 3, 4, and 5 mm for l_2, l_3, and l_4, respectively. Subsequently, seven hybrid gradient honeycomb structures were constructed, as outlined in Table 3.

2.3 Material Properties

The wheel rim was constructed using aluminum alloy (AI 7075-T6), while the two reinforcement layers were crafted from high-strength steel (ANSI 4340) [31]. The mechanical properties of these materials are given in Table 4.

Polyurethane was employed for constructing honeycomb spokes, shear bands, and inner and outer covers, while synthetic rubber was used for the tread. The hyperelastic properties of these materials were defined using the Ogden model [31]:

$$W=\sum_{i=1}^{3}\frac{2\mu_i}{\alpha_i^2}\left(\lambda_1^{\alpha_i}+\lambda_2^{\alpha_i}+\lambda_3^{\alpha_i}-3\right)+\sum_{i=1}^{3}\frac{1}{D_i}(J-1)^{2i}, \tag{10}$$

where W is the strain energy function; μ_i and α_i are temperature-dependent material parameters; λ_i is the biased main elongation; D_i is the incompressibility parameter; J is the elastic volume ratio. The specific values are shown in Table 5.

2.4 FE Analysis Methods and Mesh Convergence Analysis

To facilitate the establishment of the FE model, enhance the computational efficiency, and minimize computational cost, appropriate simplifications were applied to the NPTs. Considering computational accuracy, the primary simplifications included the following:

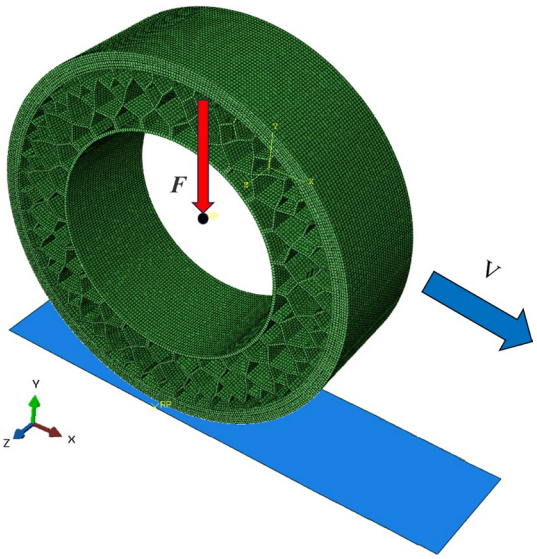


Figure 3 Finite element model of honeycomb NPTs

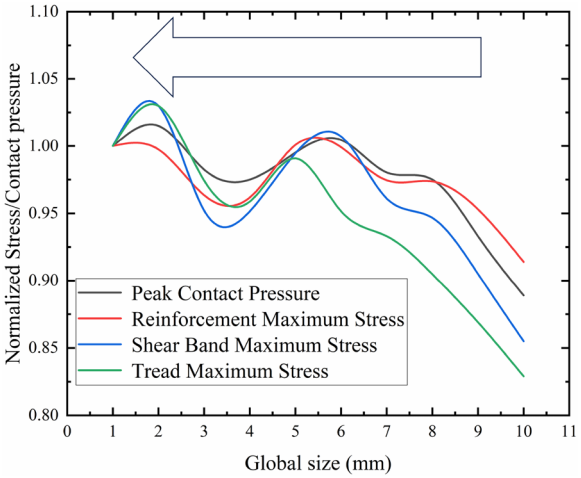


Figure 4 Mesh convergence studies

1) Disregard the friction between the contacting components.
2) Neglecting the effects of tread patterns on the performance of the NPTs.
3) Assuming that all materials are isotropic and that the tire is completely symmetrical about the wheel center plane (longitudinal-vertical) that bisects the NPT structure.
4) Ignore the influence of temperature during tire rolling.

The 3D models of the honeycomb NPTs were imported into HyperMesh software for mesh partitioning. First, two-dimensional cross-sectional meshes were created using mixed triangular/quadrilateral meshes for the honeycomb spokes and quadrilateral meshes for the remaining components. Subsequently, 3D meshes were derived using the Solid map feature of HyperMesh software. After meshing, the model was imported into ABAQUS software for simulation and analysis, with corresponding material properties assigned to each part. To establish kinematic coupling constraints, the inner rim surface was coupled to the wheel center reference point. Moreover, contact and interaction between the rigid road surface and the tread is set up, with a 0.3 friction coefficient in the tangential direction and hard contact in the direction normal to the terrain. Additionally, some wedge-shaped grids of the spokes used C3D6 elements, and the remaining Hexahedron grids employed C3D8RH simplified integration elements. The simplified FE model of the honeycomb NPTs is depicted in Figure 3.

To mitigate the effect of meshing on the FE simulation, a mesh convergence analysis was conducted (Figure 4). Figure 4 shows the dependence of the maximal local stress developed in the reinforcement, shear band and tread and peak contact pressure of the NPT-1 on the mesh size varying from 1 to 10 mm. Both the peak contact pressure and stress responses were also normalized. Through analysis, as the global mesh size decreases, the numerical calculation results fluctuate within a certain range and tend to stabilize. It can be observed that the numerical calculation results with a global mesh size of 5mm are closer to those with a global mesh size of 1mm. Considering both computational accuracy and cost, a 5 mm grid size was deemed appropriate. The NPTs with different configurations employed the same grid partitioning method, with a total node count of approximately 243700 and a total cell count of approximately 177700 for each NPT.

The simulation was conducted in two analysis steps. In the first step, the tire center reference point was completely fixed, and the rigid road was displaced upward to exert an equivalent load on each NPT model, enabling the analysis of its static performance. In the second step, the rigid road maintained the same displacement as in the first step, and an angular velocity of 26 rad/s was applied at the wheel center reference point to obtain the rolling responses of the tire in terms of maximal local stress and normal contact force or reaction force.

2.5 Experimental Verification

The support spoke structure is a key component of the NPTs. This study specifically explored the influence of the honeycomb spoke structure on the mechanical

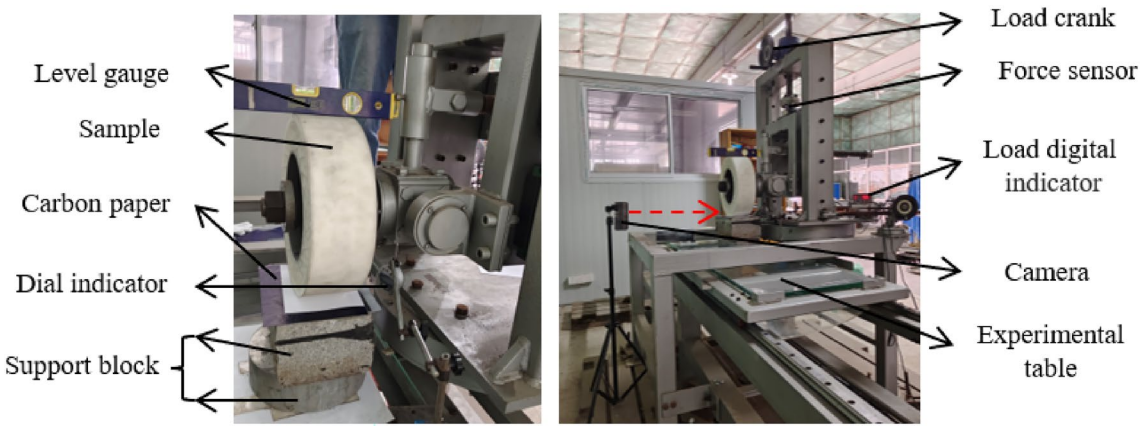


Figure 5 Static stiffness test of honeycomb NPTs

properties of the NPTs and subsequently optimized the parameters of the honeycomb spoke structure. To validate the accuracy of the numerical simulation method and the notability of the effect of gradient variation in spoke thickness on NPT performance, honeycomb NPT experimental models (EM) were fabricated using 3D printing technology, and static stiffness tests were conducted (Figure 5). To minimize manufacturing costs, a 1:2 scale model was fabricated. The rim was made of cast iron, and the rest was constructed from resin material. Two sets of static stiffness tests were conducted. The first set, EM NPT-1, had the same structure as NPT-1 in Table 3, whereas the second set, EM NPT-7, had the same structure as NPT-7 in Table 3.

The radial stiffness results obtained from numerical simulation and testing are presented in Figure 6.

In the finite element simulation results (Figure 6(a)), the radial stiffness of NPT-1 is larger than that of NPT-7. This indicates that the honeycomb structure adopted in NPT-1 is superior to that adopted in NPT-7 in terms of radial stiffness. In the experimental results (Figure 6(b)), the radial stiffness of EM NPT-1 is also larger than that of EM NPT-7. The curve results of the static stiffness test were consistent with the results of the numerical simulation test.

To further validate the accuracy of the numerical simulation results, the deformation of the test samples was examined and compared with the numerical simulation results (Figure 7). From Figure 7, a close agreement was

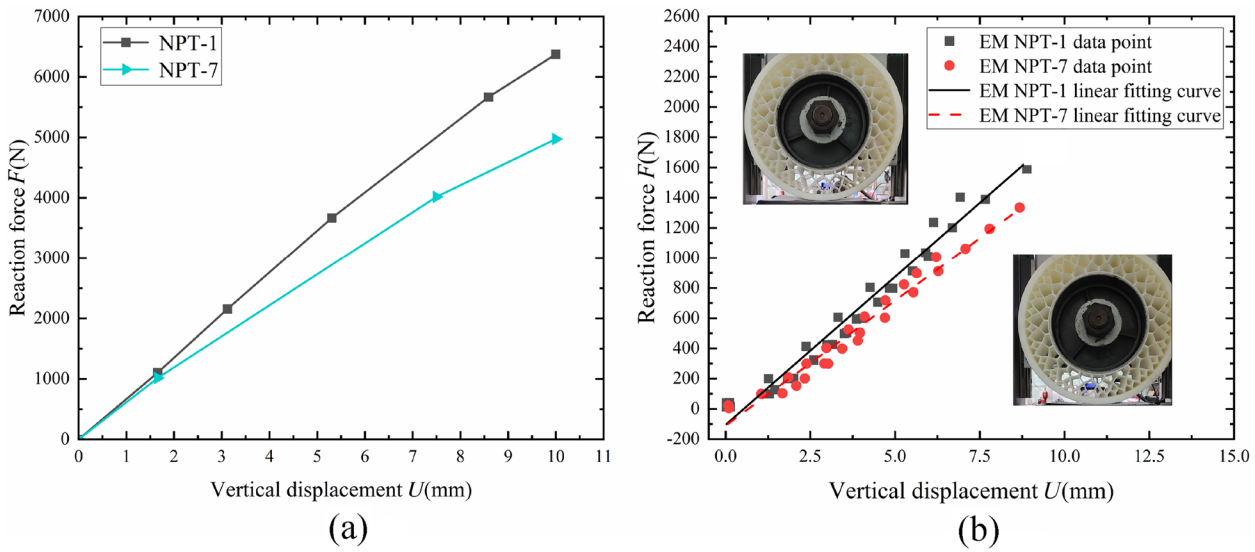
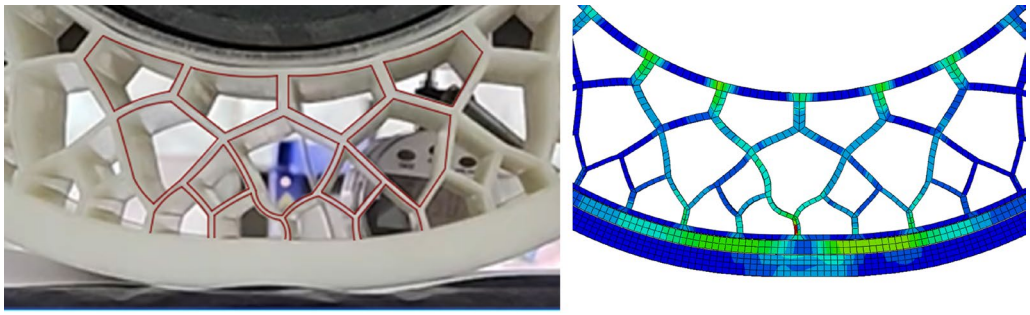


Figure 6 Radial stiffness of honeycomb NPTs: (**a**) Numerical simulation, (**b**) Test



(a) (b)

Figure 7 Comparison of deformation: (**a**) Experimental result of EM NPT-1, (**b**) Numerical simulation of NPT-1

Table 6 The applied force/displacement in different case studies

Types	Analysis items	Applied force/displacement
Static	Mesh convergence study	From 0 to 10 mm vertical displacement
	Load carrying capacity	
	Stress distribution	From 0 to 5000 N vertical force
	Deformation modes	Under 5000 N vertical force
	Contact pressure distribution	
Dynamic	Stress distribution	Under 5000 N vertical force and 26 rad/s angular velocity
	Vibration characteristics	

observed between the numerical simulation deformation results and the test deformation. The results presented in Figures 6 and 7 collectively affirm the reliability of the FE numerical simulation method.

3 Research on the Performance of Gradient Honeycomb NPTs

A comprehensive study and comparison were conducted on the static and dynamic characteristics of six mixed gradient honeycomb NPTs and one uniform thickness honeycomb NPT (Table 3). The simulation analysis of NPTs is divided into static characterization and dynamic characterization. The specific boundary conditions during the static and dynamic simulations are shown in Table 6.

3.1 Static Characteristics

In the numerical simulation of the static characteristics, motion coupling constraints were incorporated between the rim and the center reference point of the NPTs, with the center reference point fixed. Throughout the simulation, the NPTs were kept stationary, and the load was applied to the tire by applying displacement to the road surface. The contact interaction between the road surface and the tread of the NPTs was established with a friction coefficient of 0.3 in the tangential direction and a hard contact in the vertical direction.

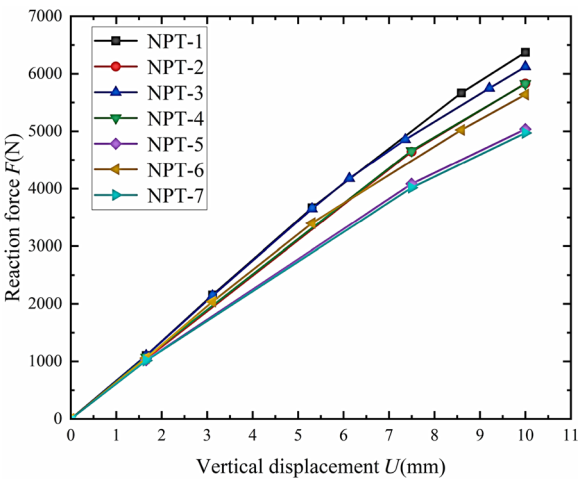


Figure 8 Radial stiffness curves of NPTs

Figure 8 illustrates the curves depicting the displacement of the road surface against the reaction force obtained in the simulation of the seven types of NPTs. These curves can be used to characterize the radial stiffness of the various NPTs. Across all NPTs, the

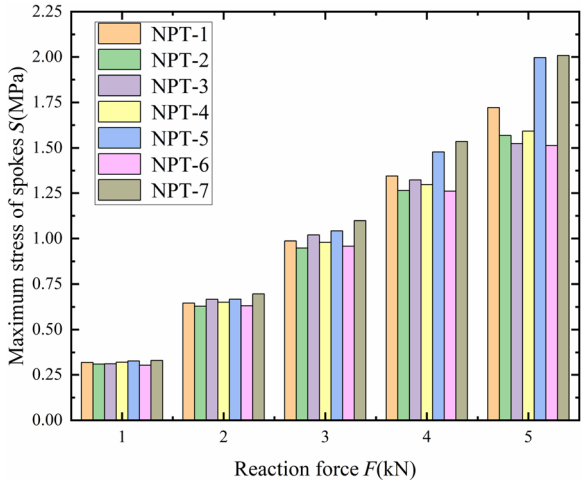


Figure 9 Variation in maximum stress of NPT spokes with load

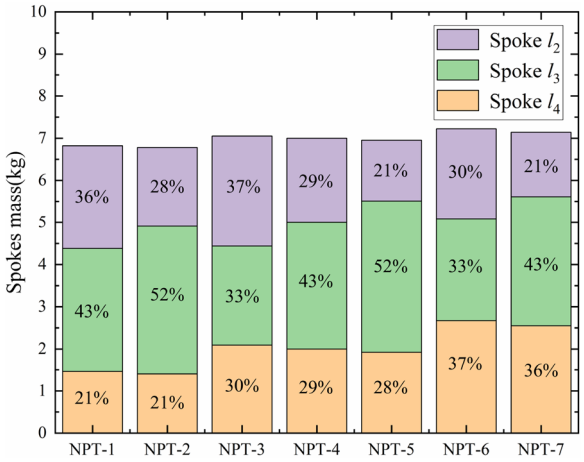


Figure 10 Mass of honeycomb spokes

reaction force on the road surface gradually increased as the displacement value of the road surface increased. When the road surface displacement reached its maximum value, NPT-1 exerted the maximum force on the road surface (indicating the greatest stiffness), whereas NPT-7 exerted the minimum force on the road surface (indicating the smallest stiffness). The curves of NPT-1 and NPT-3 appeared closely overlapped and the stiffnesses are relatively close to each other when the value of the pavement displacement is lower than 7 mm. However, as the displacement increased, the stiffness of NPT-1 surpassed that of NPT-3. The stiffness curves of NPT-5 and NPT-7 exhibited similarities throughout, but the stiffness of NPT-5 was larger than that of NPT-7. The stiffness curves of NPT-4 and NPT-2 appeared almost coincident. In summary, the stiffness of NPT-1 and NPT-3 with a spoke l_2 thickness (t_2) of 5 mm was larger than that of NPT-4, NPT-2, and NPT-6 with a t_2 of 4 mm. Meanwhile, the stiffness of NPT-4, NPT-2 and NPT-6 with a t_2 of 4 mm was also larger than that of NPT-5 and NPT-7 with a t_2 of 3 mm.

Spokes are critical components where the magnitude of stress values is a matter of concern. The variation in the maximum stress of honeycomb spokes under a load is illustrated in Figure 9. Evidently, maximum stress in the spokes was positively correlated with load. This increase in NPT-5 and NPT-7 was relatively significant, while the increase in NPT-2, NPT-3, NPT-4, and NPT-6 was relatively small. Under a load of 5000N, the stress values in the spokes of NPT-7 and NPT-6 were considerably larger than those in the other NPTs.

Since NPTs replace the pneumatic components of conventional pneumatic tires with spoke structures, NPTs exhibit higher density and greater mass. Mass, as a crucial parameter, has a significant effect on the performance of NPTs. This study focused on the effect of variations in the structural parameters of honeycomb spokes on NPT performance while maintaining the masses of the wheel rim, reinforcement layer, tread, and cover layer. The masses of different parts of honeycomb spokes are presented in Figure 10. NPT-2 had the lowest mass and NPT-6 had the highest mass. Analyzing Figure 8 in conjunction with the mass distribution, it can be observed that NPT-3 and NPT-1 exhibited higher radial stiffness, with the mass of spoke l_2 contributing a larger proportion to the overall weight of the spoke. By contrast, NPT-5 and NPT-7 exhibited lower radial stiffness, with the mass of spoke l_2 contributing a smaller proportion to the overall weight of the spoke. Additionally, the ratio of stress to weight serves as an indicator of a tire's efficiency in withstanding external forces relative to its own weight [24]. Higher ratios indicate that the structure experiences more stress for the same mass, rendering it more vulnerable to damage. Lower ratios are advantageous in the lightweighting process. Table 7 presents the ratio of maximum stress to the spoke mass (MPa/kg) for the seven NPTs when subjected to a vertical force of 5000 N. It can be clearly seen that the ratios of NPT-3 and NPT-6 are lower, which indicates that the honeycomb structures of NPT-3 and NPT-6 can withstand higher stresses with lower mass and have higher structural strength. This means that NPT-3 and NPT-6 have a better service life. The combined radial stiffness graphs highlight the superiority of the mixed gradient honeycomb structure of NPT-3 compared with the other types.

3.2 Dynamic Characteristics

In the dynamic simulation, the motion of the rim was coupled with the wheel center reference point. An

Table 7 Maximum stress and mass of NPTs spokes under 5000 N load

	NPT-1	NPT-2	NPT-3	NPT-4	NPT-5	NPT-6	NPT-7
Maximum stress of spokes (MPa)	1.7214	1.5680	1.5237	1.5922	1.9964	1.5131	2.0076
Mass of spokes (kg)	6.82	6.79	7.04	6.99	6.95	7.22	7.15
Ratio (MPa/kg)	0.2524	0.2309	0.2164	0.2278	0.2873	0.2096	0.2808

angular velocity of 26 rad/s (32.08 km/h) was set at the wheel center reference point, and the road surface was elevated a certain amount to enable the application of a load of 5000 N to the wheels. The wheels reached a stable rolling state within 1.3 s. Due to the optimal performance of NPT-3 among all NPTs with gradient structures, NPT-3 was selected to compare its dynamic characteristics with the traditional uniform structure NPT-4.

Figure 11 illustrates the time-varying reaction forces for NPT-3 and NPT-4 under steady state rolling conditions. In the case of NPT-3 and NPT-4, the normal force is observed to fluctuate around 4873.5 N and 4972.1 N, respectively. The mean value of the force during rolling was slightly less than the 5000 N force in the stationary condition, possibly due to the dynamic effect in the direction perpendicular to the road. Furthermore, the variance of the reaction force of NPT-3 and NPT-4 was 32938.73 and 36227.02, respectively. Not only was the mean reaction force of NPT-3 smaller than that of NPT-4, but the fluctuation of reaction force of NPT-3 was also less than that of NPT-4, which suggests that NPT-3 would enable a more comfortable driving experience.

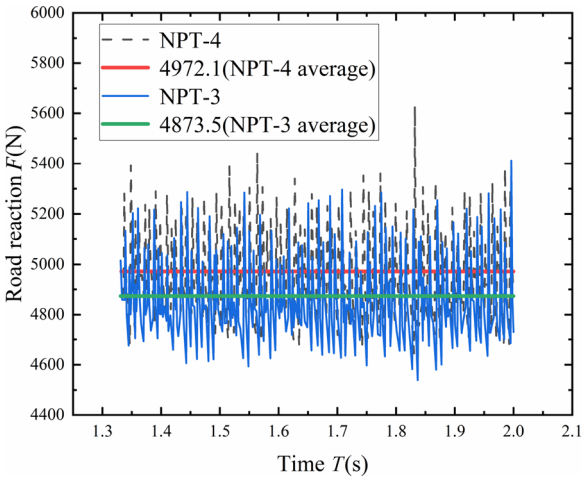


Figure 11 Time-domain variation of reaction forces for NPT-3 and NPT-4

The stresses induced in the spokes exhibited variations during the rolling of the NPTs. Figure 12 presents the stress response of spokes for NPT-3 and NPT-4 under a load of 5000 N and speed of 32.08 km/h. It can be clearly seen that the maximum stress values occurred periodically with four peaks, corresponding to the rotation period of the NPTs in the simulation time. Upon entering the compression zone, the stress at the extraction point experienced a notable surge and reached its peak value. However, a significant reduction in stress was observed as the spoke moved away from the contact region. The average spoke stresses for NPT-3 and NPT-4 were 0.2373 and 0.4314 MPa, respectively. The average stress of the NPT-3 spoke in the rolling condition was less than that of the NPT-4 spoke.

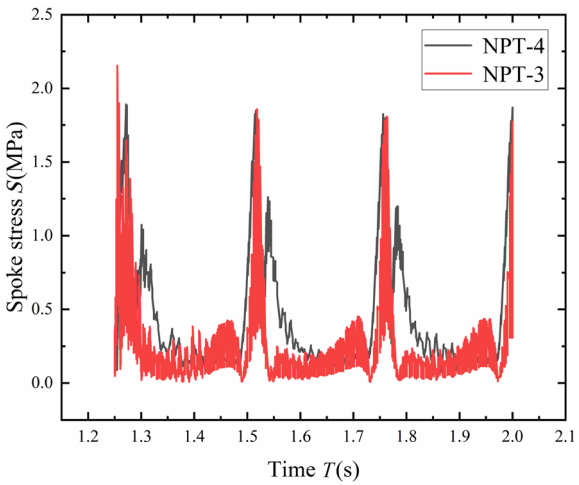


Figure 12 Time-domain variation of stress in NPT-3 and NPT-4 spokes

Overall, the NPT-3, characterized by a hybrid gradient honeycomb structure, exhibits excellent performance in both static and dynamic characteristics. It has high radial stiffness, robust structural strength, and superior vibration-damping capability. This underscores the excellent performance and promising application prospects of gradient honeycomb NPTs. Introducing functionally gradient designs into NPT design is very meaningful. The thickness parameters of honeycomb spokes, particularly thickness t_2 of spoke l_2, exert a significant effect on radial stiffness. Meanwhile, other parameters of the spokes also play an important role in influencing the characteristics of the NPTs. Therefore, it is necessary to carry out a multi-objective optimization design on this basis to

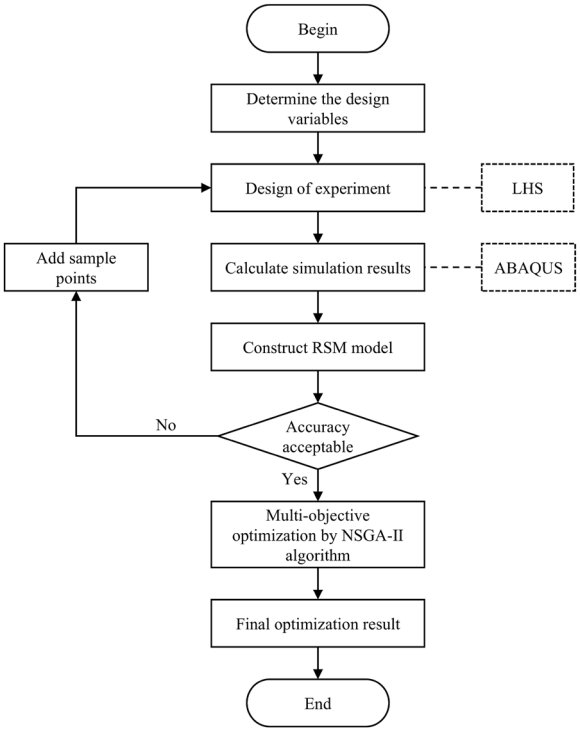


Figure 13 Multi-objective optimization process for gradient honeycomb NPTs

explore the optimal design solution for gradient honeycomb NPTs.

4 Multi-objective Optimization of Gradient Honeycomb NPTs

To obtain samples of design variables and investigate the effects of honeycomb structural parameters on radial stiffness, spoke stress, and mass of NPTs, a joint simulation using Isight software and ABAQUS software was employed after determining the design variables. Initial sample points were obtained, and a DOE was conducted. The Latin hypercube sampling (LHS) method was employed to obtain sample data. Subsequently, an approximate function solution model was constructed using an RSM with the sample data, establishing the relationship between the design variables and the corresponding output. Finally, the obtained RSM was used for multi-objective optimization. The multi-objective optimization process for gradient honeycomb NPTs in this study is presented in Figure 13.

4.1 Design Variables and DOE

The previous study concluded that variations in the thickness parameter of honeycomb spokes affect the performance of NPTs. Moreover, according to Eqs. (1)–(9), the length parameters of the spokes also exert a substantial effect on the performance of the honeycomb structure. Therefore, the selected variables included thicknesses t_2, t_3, and t_4 of the spoke l_2, l_3, and l_4, respectively, while considering projected lengths T_2 and T_3 of the spoke l_2 and l_3 in the vertical diameter direction. The initial values and ranges of these variables are detailed in Table 8. During the DOE, 90 sample points were generated using the LHS method. The LHS was first proposed by McKay et al. [32] and is considered an effective tool for enhancing efficiency in sampling methods for structural reliability analysis [33].

Table 8 Initial values and ranges of design variables

Variables	Initial value (mm)	Lower (mm)	Upper (mm)
T_2	20	5	50
T_3	30	5	50
t_2	5	2	6
t_3	4	2	6
t_4	3	2	6

4.2 Response Surface Model and Mathematical Model for Multi-objective Optimization

The use of surrogate models has garnered broad acceptance, and mathematical modeling derived from responses and variables can significantly reduce computational time. The response surface method, which was proposed by Box et al. [34], is based on reasonable DOE and the experimental data and establishes the relationship between the variable factors and the output response through fitting multiple regression equations, thereby addressing multivariate problems based on the obtained fitted function. The relationship between the objective function G and the design variables $x_1, x_2, \ldots, x_k$ is expressed in Eq. (11).

$$G = a_0 + \sum_{i=1}^{k} a_i x_i + \sum_{i=1}^{k} a_{ii} x_i^2 + \sum_{i \le j} a_{ij} x_i x_j + \varepsilon, \tag{11}$$

where a_0, a_i, a_{ii} and a_{ij} denote the intercept regression coefficients, the linear coefficients of x_i, the quadratic coefficients of x_i, and the linear interaction coefficients between x_i and x_j, respectively, and ε denotes the residuals between the actual and approximate values.

Based on the computation and analysis outcomes of the RSM, the RSM for the load-carrying capacity (RF) and the maximum stress (S) of the spokes can be written as follows:

Table 9 Accuracy of response surface model

	RF	*S*	Reference value
R^2	0.9352	0.9092	>0.9
RMSE	0.0873	0.0987	<0.2

$$RF = 127.2870 + 10.5475T_3 - 53.218t_3 - 13.9392t_2^2 - 0.1984T_3^2 + 1.7482T_2t_2 - 0.2008T_2T_3 + 1.4932T_2t_3 + 1.1295t_2T_3 + 12.3807t_2t_3 + 14.6464t_2t_4 - 0.0033T_2^3 - 0.7635t_4^3, \tag{12}$$

$$S = 2.0367 + 0.0076T_2T_3 + 0.0307T_2t_3 - 0.0277t_2T_3 - 0.4483t_2t_4 + 0.0313T_3t_3 + 0.0148T_3t_4 + 0.4996t_3t_4 - 0.0004T_2^3 + 0.0692t_2^3 - 0.0001T_3^3 - 0.0676t_3^3 - 0.0195t_4^3. \tag{13}$$

The accuracy of the approximation model can be assessed using specific standards. *R*-square(R^2), which indicates the degree of fit of the surrogate model, has been widely employed to verify the accuracy of the approximation model. Additionally, the root-mean-square error (*RMSE*) has been employed to assess the accuracy of the model. Generally, the approximate model is considered to meet the accuracy requirement when the value of R^2 exceeds 0.9 and the *RMSE* is less than 0.2. The accuracy of the *RMSE* used in this study is presented in Table 9. The relationship between the actual and predicted values of the RSM and its regression curve are shown in Figure 14. The results of Table 9 and Figure 14 indicate the predicted values closely align with the actual values. The accuracy of *RMSE* meets the requirements.

The gradient honeycomb NPTs analyzed in this study were assessed in terms of load-bearing capacity (*RF*) and maximum stress of honeycomb spokes (*S*). Additionally, the mass of NPTs was considered as an optimization objective. Since the cross-section of the NPTs remains uniform in the transverse direction and the material is homogeneous, only the structural parameters of the honeycomb spokes were modified in this study. Therefore, alterations in the mass of the NPTs result from changes in the cross-sectional area of the honeycomb spoke structure. The equivalent objective function of mass (*Mass*) can be derived from Eq. (13):

$$Mass = l_2t_2 + l_3t_3 + l_4t_4. \tag{13}$$

Honeycomb NPTs exhibit superior load-carrying capabilities and typically possess high radial stiffness. However, attention must be directed toward factors such as service life, rolling resistance and comfort. The maximum stress experienced by the honeycomb spokes plays a crucial role in fatigue damage of NPTs. Minimizing maximum stress is essential to prevent damage from excessive stress. Moreover, the mass of NPTs exerts a substantial effect on rolling resistance, with a lower mass corresponding to lower rolling resistance. Therefore, the optimization objective is to maximize the load carrying capacity (*RF*), with a weight of 0.2, and to minimize the maximum stress (*S*) and mass (*Mass*) of the honeycomb spokes, both with a weight of 0.4.

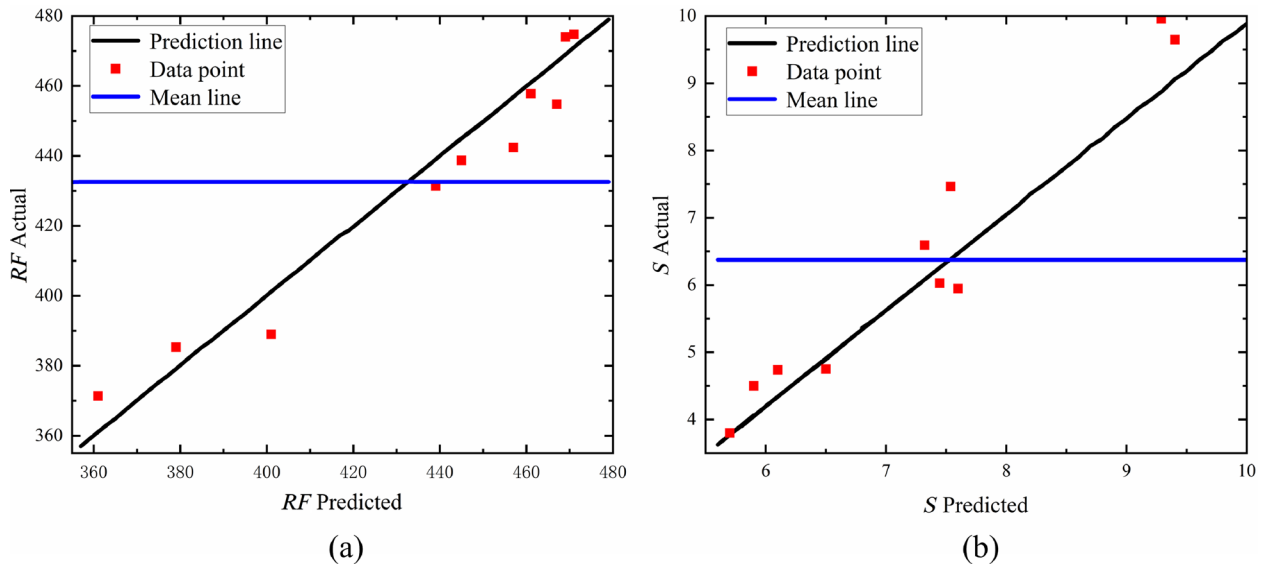


Figure 14 Comparison between the predicted curve of response surface model and actual value

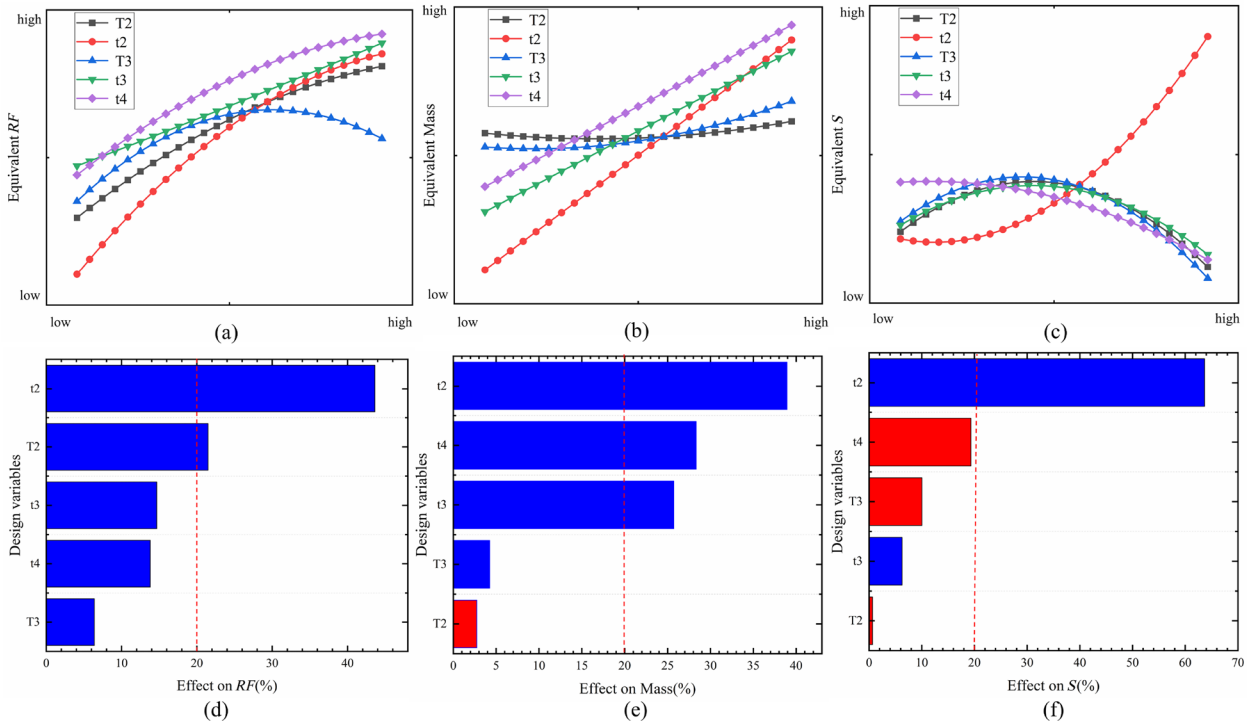


Figure 15 Main effect diagrams (**a**, **b**, **c**) and Pareto diagrams (**d**, **e**, **f**) of parameters on radial stiffness, mass, and maximum stress

$$\begin{cases} Find: T_2, T_3, t_2, t_3, t_4, \\ \max: 0.2RF; \min: 0.4S, 0.4Mass, \\ s.t. Mass \le M_0, \\ 5 \le T_2 \le 50, \\ 5 \le T_3 \le 50, \\ 2 \le t_2 \le 6, \\ 2 \le t_3 \le 6, \\ 2 \le t_4 \le 6, \\ 10 \le T_2 + T_3 \le 65, \end{cases} \tag{14}$$

where *RF*, *S*, and *Mass* represent load-carrying capacity, the maximum stress in the honeycomb spokes, and mass, respectively. M_0 denotes the initial mass of the NPT.

The multi-objective optimization algorithm used in this study is based on the Nondominated Sorting Genetic Algorithm (NSGA-II) [35]. This algorithm is characterized by its low computational complexity, and it treats each objective parameter individually. Standard genetic operations, such as mutation and crossover, were applied to the designs. The selection process involved two main mechanisms: "non-dominated sorting" and "crowded distance sorting". At the end of the optimization run, a Pareto set was formed in which each design represents the "best" combination of objective values.

4.3 Effect Analysis of the Design Variables

The radial stiffness of NPTs characterizes their load-carrying capacity and is closely related to the ride comfort of the vehicle. As NPTs undergo rolling, the spoke structure experiences alternating tensile and compressive stresses, with the maximum stress in the spoke structure serving as a key indicator parameter for assessing the service life of NPTs. The main effect plots and Pareto diagrams depicting the influence of spoke parameters on mass, radial stiffness, and maximum stress in the spoke are presented in Figure 15.

As can be seen in Figure 15(a), the radial stiffness of the NPTs increased with an increase in thicknesses (t_2, t_3, and t_4) of the honeycomb spokes and the radial projection length (T_2) of spoke l_2. The radial stiffness initially increased and then decreased with an increment of the radial projection length (T_3) of spokes l_3. In Figure 15(b), mass significantly increased with an increase of the thicknesses (t_2, t_3, and t_4) of the spokes, whereas the radial projection lengths (T_2 and T_3) of the spokes had no significant effect on mass. In the main effect plot of maximum stress of the spokes in Figure 15(c), the maximum stress first increased and then decreased with an increase of the radial projection length (T_2, T_3) and the thickness (t_3) of the spokes l_3. An increase in the thickness (t_4) of the spoke l_4 led to a decrease in maximum stress of the

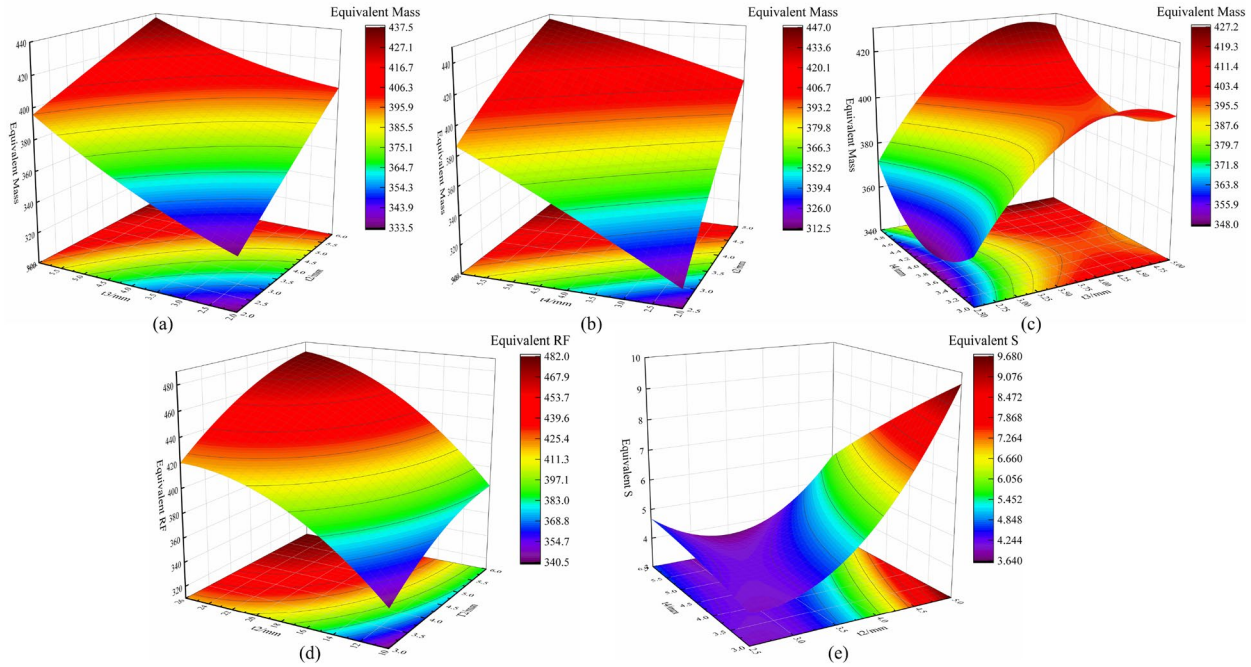


Figure 16 Effects of parameter interactions: (**a**) Interaction between t_2 and t_3 for *Mass*, (**b**) Interaction between t_2 and t_4 for *Mass*, (**c**) Interaction between t_3 and t_4 for *Mass*, (**d**) Interaction between T_2 and t_2 for *RF*, (**e**) Interaction between t_2 and t_4 for *S*

spoke, whereas an increase in the thickness (t_2) of the spoke l_2 led to an increase in the maximum stress.

In the Pareto diagram, an effect exceeding 20% of the design variables is considered significant. Figure 15(d) indicates that the effect of the thickness (t_2) of the spoke l_2 and the length (T_2) of the projection of the spoke l_2 in the radial direction on the radial stiffness of the NPTs was significant. From Figure 15(e), the thicknesses (t_2, t_4, and t_3) of the honeycomb spokes significantly influenced on mass, with the effect decreasing in that order. Figure 15(f) illustrates that the thicknesses (t_2 and t_4) of the spokes were the two variables that significantly affected the maximum stress of the spokes.

The structural parameters of the honeycomb spokes significantly affected the mass, radial stiffness and maximum spoke stress of the NPTs (Figure 15). However, interactions were also noted between the parameters. Figure 16 illustrates the interaction between the main influencing design variables for mass, radial stiffness and spoke maximum stress. As shown in Figure 16(a), the mass reached its maximum value when thicknesses (t_2 and t_3) of spokes l_2 and l_3 were at their maximum. Figure 16(b) shows that the maximum mass was achieved when the thicknesses (t_2 and t_4) of spokes l_2 and l_4 were maximum. From Figure 16(a) and (b), it can be concluded that the effect of thickness (t_2) of spoke l_2 on mass was the most significant. Figure 16(c) shows that when the thickness (t_3 and t_4) of spokes l_3 and l_4 were at their maximum, the mass was the highest; however, the effect of the thickness t_4 was not as prominent as that of the thickness t_3. Furthermore, when t_3 was large, the mass remained at a higher value. Figure 16 (d) illustrates that the radial stiffness of the NPTs was the greatest when the radial projected length (T_2) of the spoke l_2 and the thickness t_2 were the highest, with the thickness t_2 exerting a more pronounced effect on the radial stiffness. Figure 16(e) shows the effect of spoke thicknesses t_2 and t_4 on the maximum stress of the spokes. When the value of t_4 was the minimum and that of t_2 was maximum, the spoke stress value was maximum. However, the effect of thickness t_4 on spoke stress was not significant.

4.4 Optimization Results

Based on the obtained RSM, NSGA-II algorithm was selected to solve the three-objective optimization problem. Following multi-objective optimization, a Pareto-optimal solution set was obtained (Figure 17). The set indicates that achieving the optimum values for all three objectives (mass, radial stiffness, and maximum stress of spokes) simultaneously is challenging. To identify the trade-off optimal solution among the three objectives from the Pareto solution set, the three objectives should be selected based on their weights (Eq. (14)). As the radial stiffness of honeycomb NPTs is generally high, the considerations of service life, rolling resistance, and comfort should be emphasized. The final optimized design

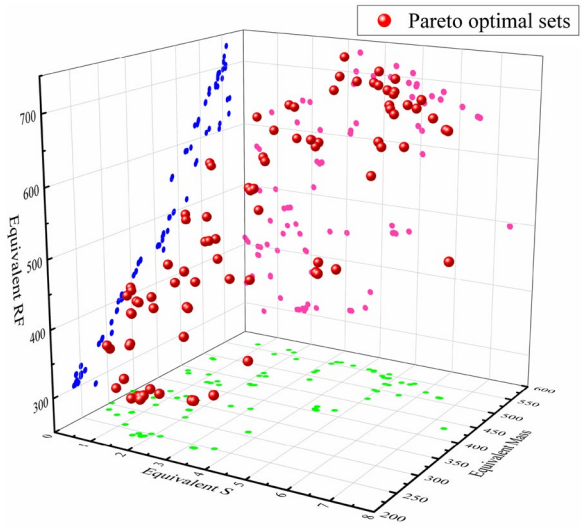


Figure 17 Distribution of Pareto-optimal solutions

Table 10 Comparison of parameters and performance of NPTs before and after optimization

Parameters	NPT-1	NPT-3	NPT-4	NPT-OP
T_2	20	20	20	27.37
T_3	30	30	30	24.90
t_2	5	5	4	3.15
t_3	4	3	4	2.14
t_4	3	4	4	3.48
Load carrying capacity (N/mm)	637.44	612.24	582.80	487.79
Maximum stress of spokes (MPa)	1.7214	1.5237	1.5922	1.5674
Mass of spokes (kg)	6.82	7.04	6.99	5.67
Ratio (MPa/kg)	0.2524	0.2164	0.2278	0.2764

parameters obtained are presented in Table 10. The specific structural differences between NPT-OP and other NPTs are shown in Figure 18.

The static and dynamic characteristics of the optimized NPT (NPT-OP) were investigated using ABAQUS software, and the results were compared with the performance of NPT-1, NPT-3, and NPT-4. The static characteristics, as outlined in Table 10. To verify the optimization method and the accuracy of the optimization results, the experimental models EM NPT-OP and EM NPT-4 were built by using 3D printing technology and static stiffness experiments were carried out. To compare the difference in radial stiffness between NPTs and pneumatic tires. Radial stiffness experiments of pneumatic tires are also carried out. The simulation and experimental results are shown in Figure 19. Both simulation and experimental results show a significant reduction in radial stiffness (load-carrying capacity) of the NPT-OP compared with the other NPTs. Meanwhile, both simulation and experimental results show that the radial stiffnesses from high to low are NPT-1, NPT-4, NPT-7, and NPT-OP. The simulation results are consistent with the experimental results, verifying the accuracy of the finite element method. The effectiveness of the optimization method and the accuracy of the optimization results are also verified. However, Figure 19(a) shows the static stiffness of NPT-OP is still greater than that of conventional pneumatic tire. This ensures the load carrying capacity of the NPTs. Therefore, taking the load-bearing capacity of the tire as a reference, the radial stiffness of the optimized NPT-OP in this paper is still within a reasonable range. Additionally, the maximum spoke stress of the NPT-OP was substantially lower than that of NPT-1 and NPT-4, which is beneficial for service life. Although the ratio of maximum spoke stress to spoke mass of the NPT-OP was the highest, this is a result of the substantial reduction in spoke mass. The spoke mass of the NPT-OP exhibited a reduction of more than 16.9% compared with the other NPTs, contributing considerably to saving energy consumption.

The contact characteristics of tires, being the only component of the vehicle in direct contact with the road surface, play a pivotal role in tire performance. The contact pressure distribution of NPT-3 and NPT-OP under a 5000 N load is presented in Figure 20. The peak contact pressure of NPT-OP was smaller than that of NPT-3, and the contact area of NPT-OP was larger than that of NPT-3. The reduced peak contact pressure is beneficial for improving wear resistance, and the expanded contact area contributes to the acceleration and braking performance of the tire.

The time-varying plots of road surface forces for NPT-3 and NPT-OP during steady state rolling under a load of 5000 N are presented in Figure 21. The data in Figure 21 reveal that the mean road surface force of NPT-OP was smaller than that of NPT-3. The results of the road surface force variance in Table 11 further emphasize that the fluctuation of the road surface force of NPT-OP was minimized. Moreover, NPT-OP not only exhibited a smaller mean road surface force but also demonstrated reduced force fluctuation. This indicates that NPT-OP has superior energy absorption and vibration-damping performance.

To investigate the reasons underlying the energy-absorbing and vibration-damping effect of NPT-OP, Figure 22 depicts the variation of the contact imprints of NPT-3 and NPT-OP during the rolling process. Additionally, Figure 23 presents the distribution of contact pressure along the peak grounding pressure line (black dashed line in Figure 22). The observed changes in Figure 22 follow a periodical pattern during the steady-state

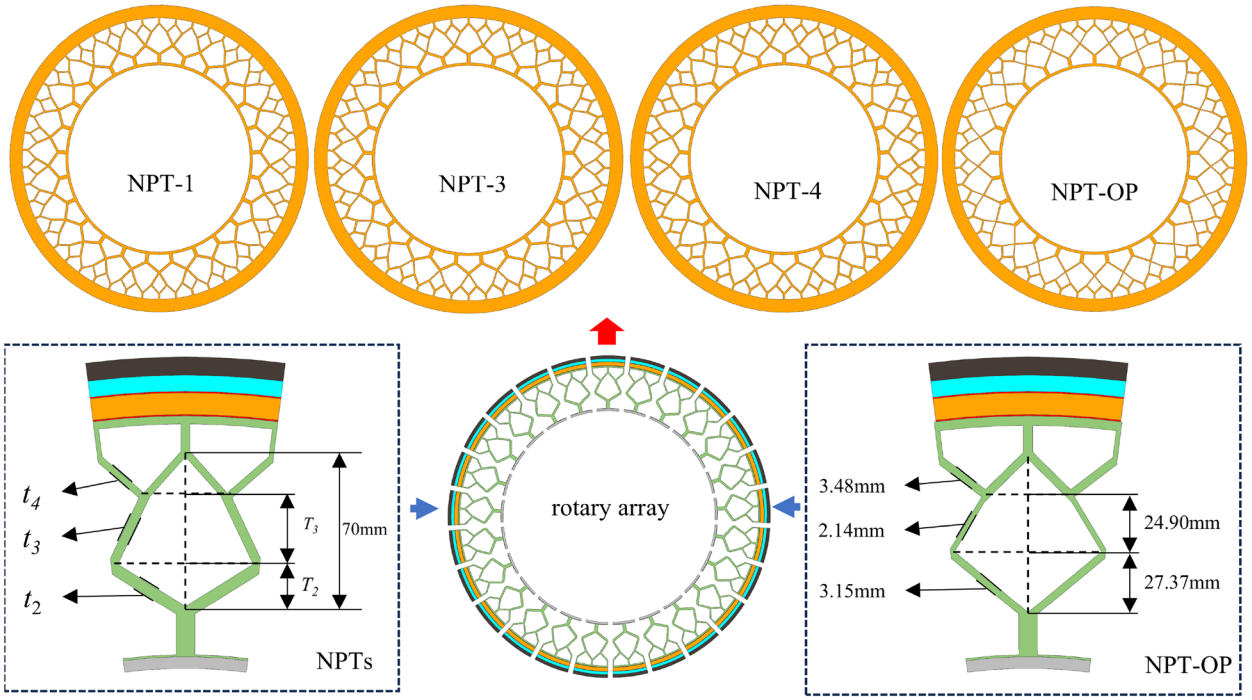


Figure 18 Specific differences between NPT-OP and NPTs

rolling process. The comparison revealed that the contact area of NPT-OP was larger than that of NPT-3, whereas the peak contact pressure of NPT-OP was smaller than that of NPT-3. The variance of the pressure values on the nodes was also calculated, and the results revealed that the variance of the contact pressure of NPT-OP was smaller than that of NPT-3, indicating that the distribution of the contact pressure of NPT-OP was more uniform than that of NPT-3. Figure 22 indicates that the maximum contact pressure gradually decreased from moment T to T + 3Δt, and the maximum contact pressure of NPT-OP was smaller than that of NPT-3 at the same moment. This finding suggests that the vibration-damping effect of NPT-OP was attributed to the dynamic change of contact pressure and the reduction of the peak contact pressure values.

Figure 24 presents the stress responses of NPT-3 and NPT-OP spokes under a load of 5000 N and a speed of 32.08 km/h. The stress peaks for NPT-3 were higher than those of NPT-OP, whereas the minimum spoke stress for NPT-3 was lower than that of NPT-OP. This suggests that NPT-OP exhibits a smaller range of stress variation, and the variance of the spoke stress in Table 11 also confirmed this result. Table 11 further shows that the peak stress of NPT-OP was the lowest, indicating the effectiveness of the multi-objective optimization in reducing spoke stress.

5 Conclusions

This paper introduces the concept of functional gradient into the structural design of honeycomb NPTs and proposes a honeycomb NPT with a thickness gradient. The study focuses on investigating the static and dynamic characteristics of this type NPTs, emphasizing load-carrying capacity, maximum stress of spokes, and vibration-damping performance. Parametric modeling of the gradient honeycomb NPTs was conducted to explore the optimal structural design. The use of a multi-objective optimization method enhanced the performance of the gradient honeycomb NPTs. The theoretical foundation and technical methods presented in this paper contribute to the advancement of structural design and optimization for gradient honeycomb NPTs. The conclusions of this paper are as follows:

(1) The structural parameters of honeycomb spokes significantly influence NPT performance. The thickness t_2 of the spoke l_2 is the primary influencing factor for radial stiffness, spoke stress and mass, and is the key design parameter of honeycomb NPTs. Mass is mainly affected by spoke thicknesses t_2, t_4 and t_3, and the maximum spoke stress is closely related to spoke thicknesses t_2 and t_4. An increase in t_2 enhances radial stiffness and reduces maximum spoke stress but may hinder lightweighting.

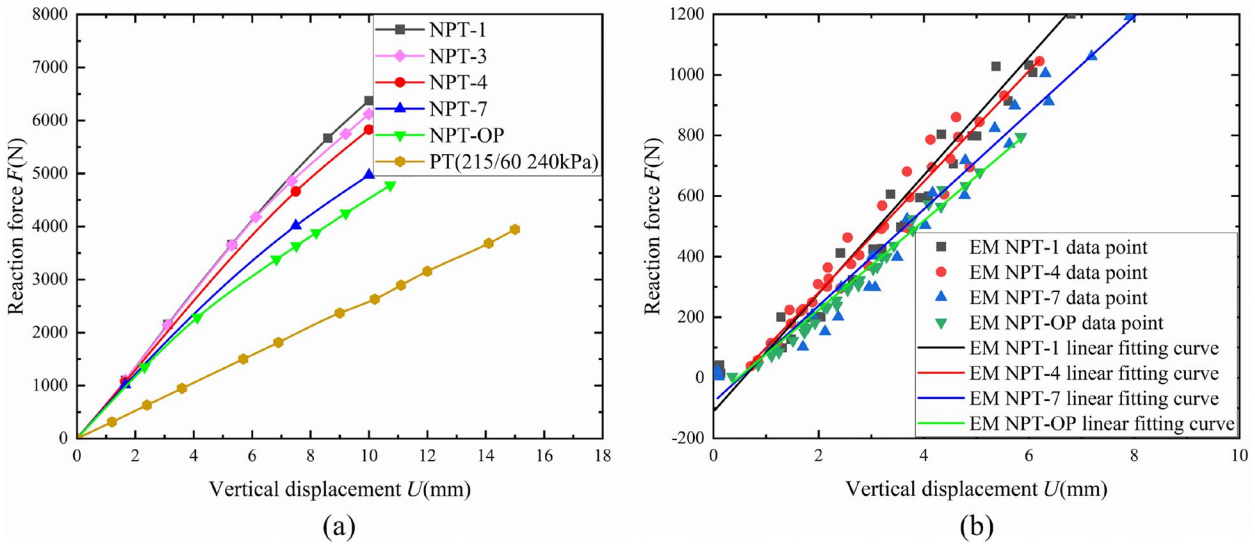


Figure 19 Comparison of radial stiffness (PT is pneumatic tire): (**a**) Numerical simulation, (**b**) Testing

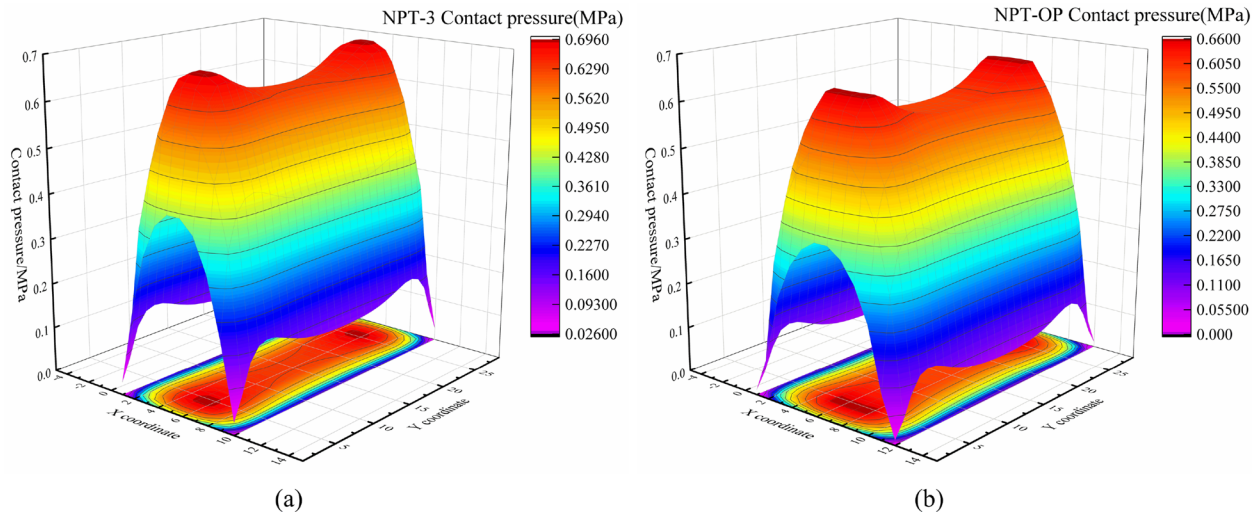


Figure 20 Contact pressure distribution for (**a**) NPT-3 and (**b**) NPT-OP

Table 11 Comparison of dynamic performance of NPTs before and after optimization

Parameters	NPT-1	NPT-3	NPT-4	NPT-OP
Average road force (N)	4934.086	4873.494	4972.107	4796.472
road force variance	35276.41	32938.73	36227.02	23609.32
Maximum spoke stress (MPa)	2.42432	1.85990	1.87067	1.33939
Minimum spoke stress (MPa)	0.010009	0.006617	0.074489	0.054835
Spoke stress variance	0.118398	0.078871	0.166770	0.036994

(2) A well-designed gradient honeycomb structure is beneficial for enhancing NPT performance. The radial stiffness of NPT-1 and NPT-3, which had gradient structure, was higher than that of NPT-4, which had uniform thickness. NPT-3 also demonstrated lower maximum spoke stress compared with NPT-4. Dynamic analysis revealed that NPT-3 offers superior ride comfort, with smaller average and variance values of the reaction force on the road surface. Additionally, compared with NPT-4, NPT-3 exhibited a smaller stress peak in the spokes, contributing to improving fatigue life of the NPTs.

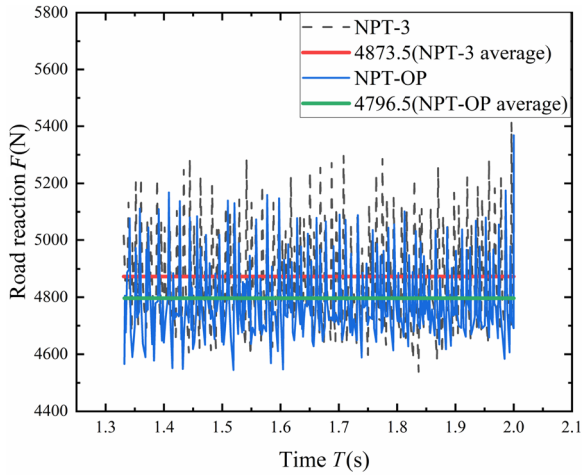


Figure 21 Time-domain variation of NPT-3 and NPT-OP reaction forces

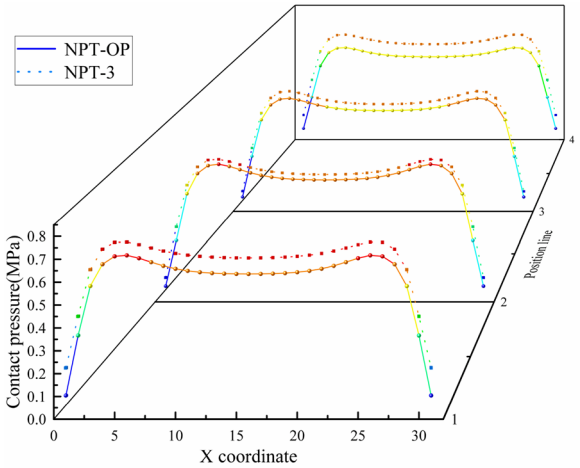


Figure 23 Peak contact imprint in NPT-3 and NPT-OP

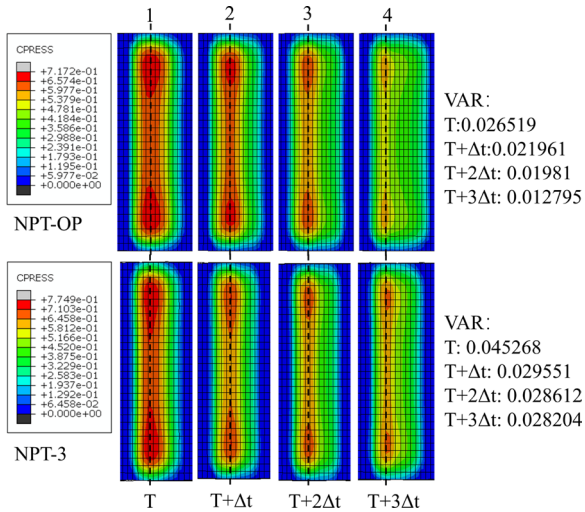


Figure 22 Contact pressure variations in NPT-3 and NPT-OP

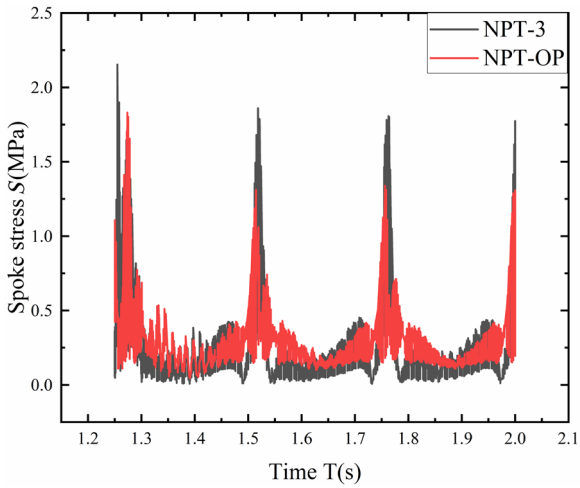


Figure 24 Time-domain stress variation in spokes of NPT-3 and NPT-OP

(3) Multi-objective optimization resulted in an NPT-OP with a spoke thickness adopting a gradient structure similar to that of NPT-3, leading to an appropriate reduction in radial stiffness. In the static analysis with the same load, the maximum stress of the spokes of the NPT-OP was reduced by 8.95% compared with NPT-1, along with a spoke mass reduction exceeding 16.9% compared with the other NPTs. In dynamic analysis with the same load, the peak spoke stress of the NPT-OP was reduced by more than 27.99% compared to the other NPTs, coupled with significantly reduced mean road surface force and lower force and stress variances. The multi-objective optimization approach proved effective for optimizing NPT performance and promoting lightweight design.

Acknowledgements
Not applicable.

Author Contributions
HZ and GL were in charge of the whole trial; HC and HF wrote the manuscript and were in charge of the whole analysis; ZZ revised the final manuscript. All authors read and approved the final manuscript.

Funding
Supported by National Natural Science Foundation of China (Grant Nos. 52072156, 52272366), Postdoctoral Foundation of China(Grant No. 2020M682269).

Data availability
The datasets generated during and/or analyzed during the current study are available from the corresponding author upon reasonable request.

Declarations

Competing Interests
The authors declare no competing financial interests.

Received: 29 March 2024 Revised: 6 April 2025 Accepted: 15 April 2025
Published online: 26 May 2025

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